



Parkinson's Disease Dataset Analysis

The purpose of this research paper is to provide a comprehensive, step-by-step guide for conducting **Descriptive Statistics** and **ANOVA's analysis** using the Isalos Analytics Platform. Descriptive statistics summarise the main characteristics of the patient cohort.

These methods are fundamental in healthcare research for identifying demographic and lifestyle factors associated with disease severity, cognitive performance, and functional impairment.

Isalos version used: 2.1.5

The dataset used in this analysis is the **Parkinson's Disease Dataset Analysis**, available on Kaggle.

Source: [Parkinson's Disease Dataset Analysis](#)

This dataset simulates real-world healthcare records and includes comprehensive information for **2,105** patients diagnosed with Parkinson's Disease. It contains demographic details (age, gender, ethnicity, education), lifestyle factors (smoking, alcohol consumption, physical activity, diet quality), and clinical measurements (UPDRS, MoCA, Functional Assessment). It is designed to support predictive modelling and risk assessment for Parkinson's disease progression and cognitive decline.

Step 1: Import Data from file

Firstly, launch the Isalos Analytics Platform (version 2.1.2). Then, in the left spreadsheet panel, create a new tab by clicking **File** and choose **Open** to load the Parkinson's Disease dataset (or simply copy-paste the data from your spreadsheet).

Col1	Col2 (I)	Col3 (I)	Col4 (I)	Col5 (I)	Col6 (D)	Col7 (I)	Col8 (D)	Col9 (D)	Col10 (D)
User Row ID	Age	Gender	Ethnicity	EducationLevel	BMI	Smoking	AlcoholConsumption	PhysicalActivity	DietQuality
3058	85	0	3	1	19.61987796	0	5.108240607	1.380659917	3.893969135
3059	75	0	0	2	16.24733916	1	6.027648029	8.40980405	8.51342825
3060	70	1	0	0	15.36823871	0	2.242135331	0.213274591	6.498804606
3061	52	0	0	0	15.45455733	0	5.997787563	1.375045164	6.715033333
3062	87	0	0	1	18.61604177	0	9.775242923	1.188607062	4.657572037
3063	68	1	2	1	39.42331141	1	13.5968889	7.796704004	7.070238878
3064	78	1	0	0	30.54200329	1	2.011281313	9.028536304	9.838445926
3065	70	1	0	0	36.75828161	1	19.98886597	3.891748622	3.421960006
3066	80	0	2	1	22.3805865	1	7.293287715	2.595670177	4.784827139
3067	71	0	3	2	23.72708628	1	17.78290985	7.344890316	3.393018462
3068	70	0	0	3	38.48254474	0	6.639761993	7.872186697	9.225710184
3069	53	0	0	1	35.8967387	1	5.212906263	7.185203018	7.918912222
3070	74	0	2	2	30.22551208	0	3.762918066	4.316651243	5.112520425
3071	87	1	1	2	38.29830655	0	12.61599522	9.299289996	6.71557911
3072	58	0	0	3	34.96532287	1	11.70859735	4.392463152	5.182038871
3073	56	1	0	0	18.95878125	1	2.04712047	9.432830343	1.720277262
3074	54	1	0	0	28.04958007	1	7.260733832	4.311617423	2.509936233

Step 2: Access the Descriptive Statistics and ANOVA functions

Navigate to the descriptive statistics module through the main menu by choosing:

Statistics → Basic Statistics → Descriptive Statistics

Then you will see the descriptive statistics configuration window.


Descriptive Statistics
?
×

Select Type
Sample

1. Sample Size & Counts
☐ Size/Count (n)
☐ Frequency
☐ Sum (Σ)
☐ Sum of Squares (SS)

2. Central Tendency
☐ Mean ($\bar{x}$)
☐ Median ($\tilde{x}$)
☐ Mode
☐ Midrange (MR)

3. Dispersion / Spread
☐ Standard Deviation (s)
☐ Variance (s^2)
☐ Range (R)
☐ Mean Absolute Deviation (MAD)
☐ Minimum (min)
☐ Maximum (max)

4. Quartiles, UQR & Outliers
Method: ☐ Inclusive ☐ Exclusive
☐ Quartiles (Q1, Q2, Q3)
☐ Interquartile Range (IQR)
☐ Outliers

5. Relative / Advanced Measures
☐ Coefficient of Variation (CV)
☐ Relative Standard Deviation (RSD)
☐ Root Mean Square (RMS)
☐ Standard Error of the Mean (SE $\bar{x}$)

6. Distribution Shape
☐ Skewness (γ_1)
☐ Kurtosis (β_2)
☐ Kurtosis Excess (α_2)

7. Normality Tests
☐ Shapiro-Wilk
☐ Kolmogorov-Smirnov
☐ Anderson-Darling

8. Plots
☐ Q-Q Plot
☐ Histogram
☐ Box Plot
☐ Violin Plot

Execute
Cancel

Step 3: Configure parameters

In the descriptive statistics window, set the following parameters:

1. Sample Size & Counts
 - a. Size / Count (n)
 - b. Frequency
 - c. Sum
 - d. Sum of Squares
2. Central Tendency
 - a. Mean
 - b. Median
 - c. Mode
 - d. Midrange
3. Dispersion / Spread
 - a. Standard Deviation
 - b. Variance
 - c. Range
 - d. Mean Absolute Deviation
 - e. Minimum

- f. Maximum
- 4. Quartiles, UQE & Outliers
 - a. Quartiles (Q1,Q2,Q3)
 - b. Interquartile Range (IQR)
 - c. Outliers
- 5. Relative / Advanced Measures
 - a. Coefficient of Variation
 - b. Relative Standard Deviation
 - c. Root Mean Square
 - d. Standard Error of the Mean
- 6. Distribution Shape
 - a. Skewness
 - b. Kurtosis
 - c. Kurtosis excess
- 7. Normality Tests
 - a. Shapiro-Wilk
 - b. Kolmogorov-Smirnov
 - c. Anderson-Darling
- 8. Plots
 - a. Q-Q Plot
 - b. Histogram
 - c. Box Plot
 - d. Violin Plot

The above preview will display the multivariate model as shown below:


Descriptive Statistics
?
×

Select Type
Sample

1. Sample Size & Counts

- ☒ Size/Count (n)
- ☒ Frequency
- ☒ Sum (Σ)
- ☒ Sum of Squares (SS)

2. Central Tendency

- ☒ Mean ($\bar{x}$)
- ☒ Median ($\tilde{x}$)
- ☒ Mode
- ☒ Midrange (MR)

3. Dispersion / Spread

- ☒ Standard Deviation (s)
- ☒ Variance (s^2)
- ☒ Range (R)
- ☒ Mean Absolute Deviation (MAD)
- ☒ Minimum (min)
- ☒ Maximum (max)

4. Quartiles, UQR & Outliers
Method: ☒ Inclusive ☒ Exclusive

- ☒ Quartiles (Q1, Q2, Q3)
- ☒ Interquartile Range (IQR)
- ☒ Outliers

5. Relative / Advanced Measures

- ☒ Coefficient of Variation (CV)
- ☒ Relative Standard Deviation (RSD)
- ☒ Root Mean Square (RMS)
- ☒ Standard Error of the Mean (SEX)

6. Distribution Shape

- ☒ Skewness (γ_1)
- ☒ Kurtosis (β_2)
- ☒ Kurtosis Excess (α_2)

7. Normality Tests

- ☒ Shapiro-Wilk
- ☒ Kolmogorov-Smirnov
- ☒ Anderson-Darling

8. Plots

- ☒ Q-Q Plot
- ☒ Histogram
- ☒ Box Plot
- ☒ Violin Plot

Distribution: Normal

Execute

Cancel

After configuring all parameters:

- * Review all selections to ensure accuracy.
- * Click the **Execute** button to run the descriptive analysis.
- * Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

Factors	Size/Count	Sum	Sum of Squares	Mean	Median	Mode	Mid Range	Standard Deviation	Variance
Age	2105.0	146512.0	282846.3923990	69.6019002	70.0	84.0	69.5	11.5945115	134.4326960

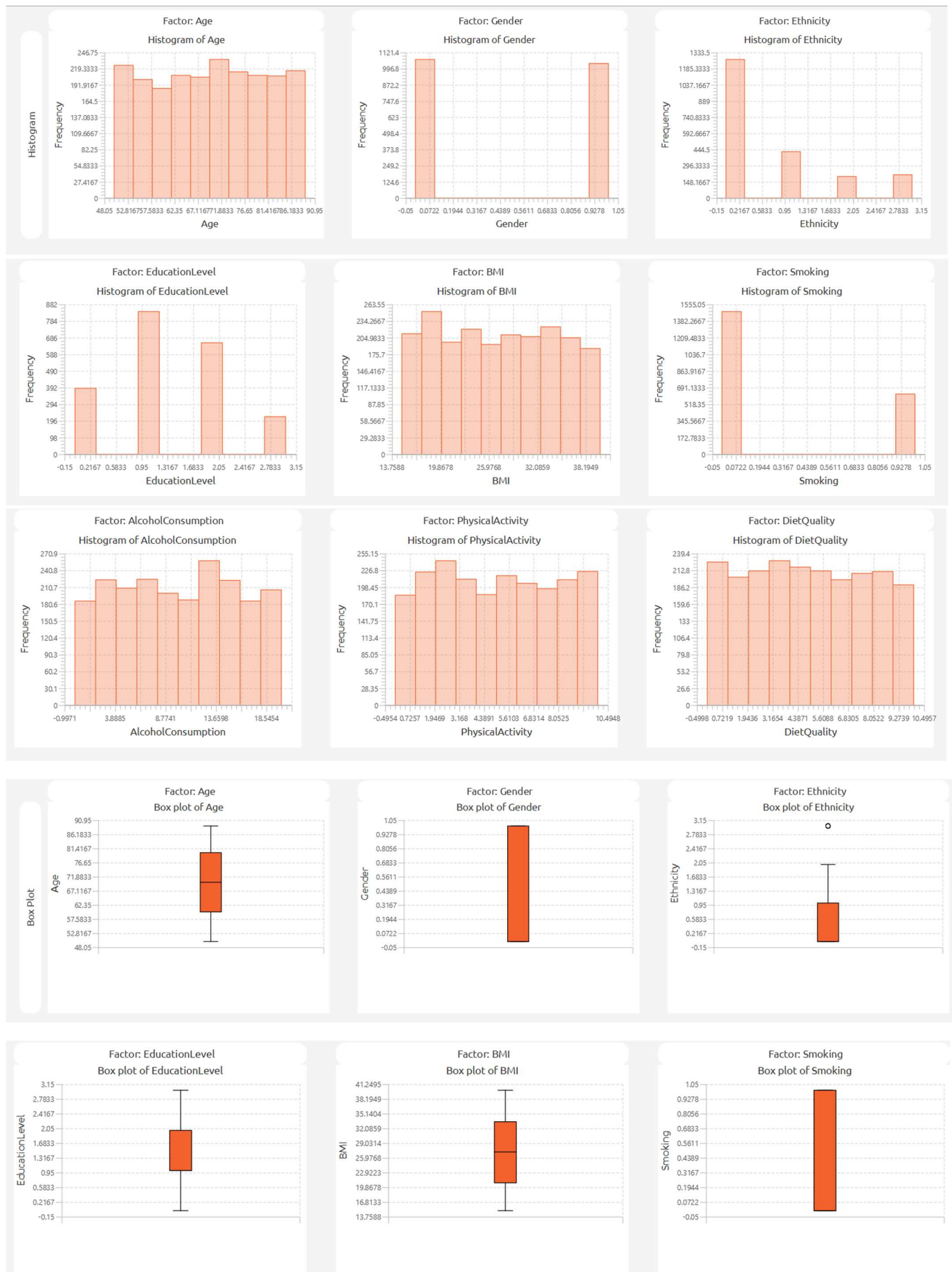
Range	Mean Absolute Deviation	Minimum	Maximum	Quartiles - Inclusive	Quartiles - Exclusive	Interquartile Range - Inclusive	Interquartile Range - Exclusive
39.0	9.9904492	50.0	89.0	Q1: 60.000000, Q2: 70.000000, Q3: 80.000000	Q1: 60.000000, Q2: 70.000000, Q3: 80.000000	20.0	20.0

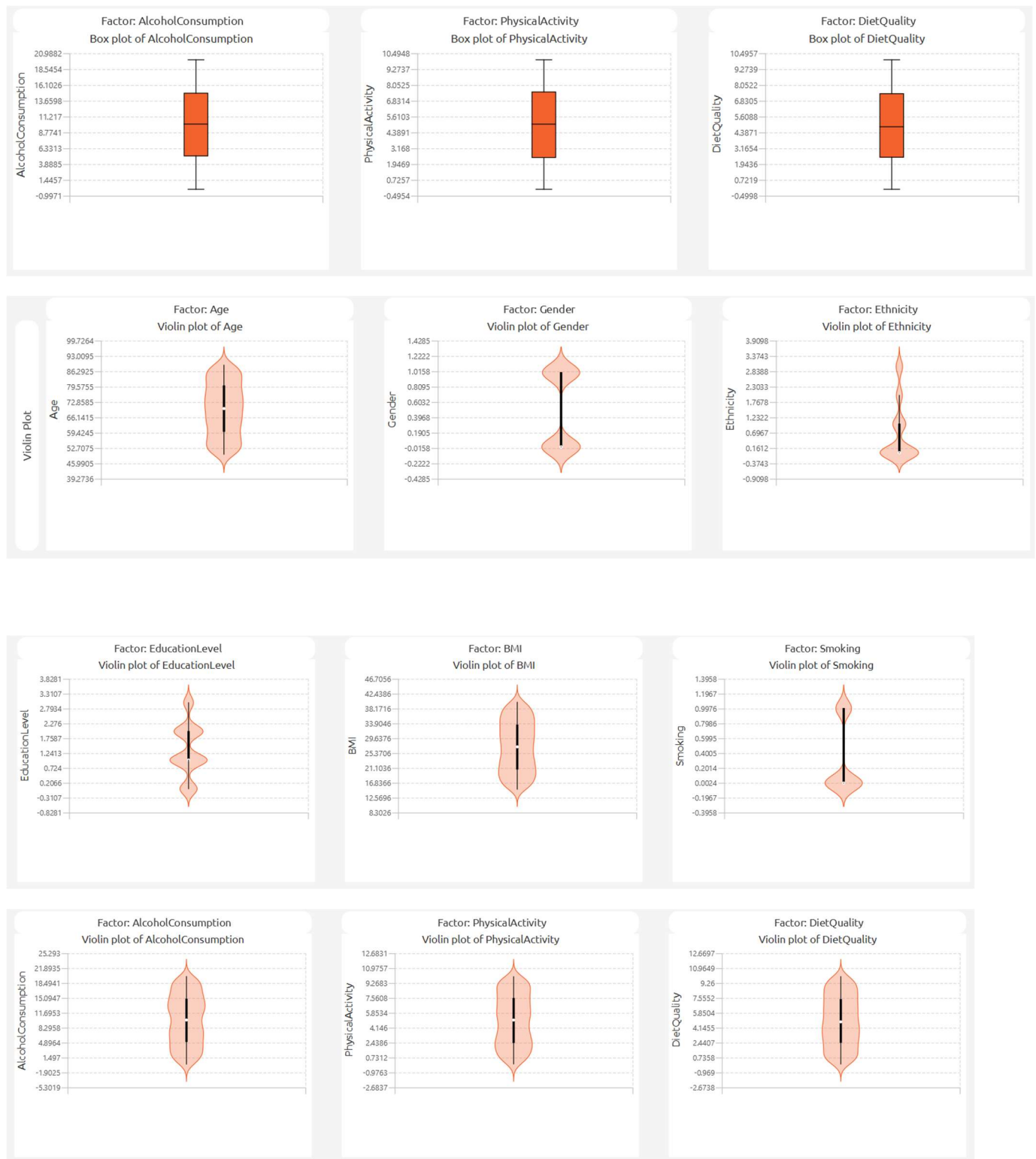
Coefficient Of Variation	Relative Standard Deviation	Root Mean Square	Standard Error Of Mean	Skewness	Kurtosis	Kurtosis Excess	Shapiro-Wilk: W	Shapiro-Wilk: p-value	Shapiro-Wilk: Prob
0.1665833	16.6583260	70.560565	0.2527123	-0.0334598	1.8160222	-1.1839778	0.9534394	0.0	1.0

Shapiro-Wilk: p-value	Shapiro-Wilk: Prob	Shapiro-Wilk: Decision	Anderson-Darling: A ²	Anderson-Darling: p-value	Anderson-Darling: Prob	Anderson-Darling: Decision	Kolmogorov-Smirnov: D	Kolmogorov-Smirnov: p-value	Kolmogorov-Smirnov: Prob
0.0	1.0	Reject Null Hypothesis	0.0691501	0.0	1.0	Reject Null Hypothesis	23.2069780	0E-7	1.0

Shapiro-Wilk: Decision	Anderson-Darling: A ²	Anderson-Darling: p-value	Anderson-Darling: Prob	Anderson-Darling: Decision	Kolmogorov-Smirnov: D	Kolmogorov-Smirnov: p-value	Kolmogorov-Smirnov: Prob	Kolmogorov-Smirnov: Decision
Reject Null Hypothesis	0.0691501	0.0	1.0	Reject Null Hypothesis	23.2069780	0E-7	1.0	Reject Null Hypothesis







Part 2: ANOVA ANALYSIS

Step 1: Import Data

If you followed the previous descriptive statistics guide, your dataset is already loaded in Isalos. Ensure the data is visible in the left spreadsheet panel.

Step 2: Access the ANCOVA function

Navigate to the ANCOVA analysis module through the main menu by choosing:

Statistics → Analysis of (Co)Variance → ANCOVA

The ANCOVA configuration window will appear:

The ANCOVA configuration window is shown with the following settings:

- Confidence Level (%): Double (0,100), Default: -
- Dependent Variable: (Empty dropdown)
- Sum of Squares for Tests: Adjusted (Type III)
- Coding for Factors: (-1, 0, +1)
- Excluded Columns: Col2 -- PatientID, Col3 -- Age, Col4 -- Gender, Col5 -- Ethnicity, Col6 -- EducationLevel, Col7 -- BMI, Col8 -- Smoking, Col9 -- AlcoholConsumption, Col10 -- PhysicalActivity, Col11 -- DietQuality
- Factors: (Empty box)
- Covariates: (Empty box)
- Formula Type: ☒ Custom, ☐ Main Effects, ☐ Full Factorial
- Formula: (Empty text box)
- Buttons: Execute, Cancel

Step 3: Configure ANCOVA parameters

The ANCOVA model we will estimate is:

$$BMI = \beta_0 + \beta_1(Age) + \beta_2(Gender) + \beta_3(Smoking) + \beta_4(Gender \times Smoking) + \varepsilon$$

This model predicts BMI (continuous) based on the covariate Age, the two categorical factors (Gender and Smoking), and their **interaction**.

In the ANCOVA configuration window, fill in the fields exactly as follows:

The ANCOVA configuration window is shown with the following settings:

- Confidence Level (%)**: 95
- Dependent Variable**: Col7 -- BMI
- Sum of Squares for Tests**: Adjusted (Type III)
- Coding for Factors**: (-1, 0, +1)

Excluded Columns (highlighted with an orange box):

- Col2 -- PatientID
- Col6 -- EducationLevel
- Col9 -- AlcoholConsumption
- Col10 -- PhysicalActivity
- Col11 -- DietQuality
- Col5 -- Ethnicity

Factors (highlighted with a blue box):

- Col4 -- Gender
- Col8 -- Smoking

Covariates (highlighted with a blue box):

- Col3 -- Age

Formula Type: ☒ Full Factorial

Formula: Age + Gender + Smoking + Age:Gender + Age:Smoking + Gender:Smoking +

Buttons: Execute, Cancel

After configuring all parameters:

- * Review all selections to ensure accuracy.
- * Click the **Execute** button to run the descriptive analysis.

- * Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant
Age	1	124.0047029	124.0047029	2.3957645	0.1218152	No
Gender	1	1.5833659	1.5833659	0.0305905	0.8611739	No
Smoking	1	1.9316053	1.9316053	0.0373185	0.8468367	No
Age*Gender	1	8.1105965	8.1105965	0.1566963	0.6922568	No
Age*Smoking	1	1.4292327	1.4292327	0.0276127	0.8680386	No
Gender*Smoking	1	3.3016479	3.3016479	0.0637877	0.8006313	No
Age*Gender*Smoking	1	9.2546383	9.2546383	0.1787991	0.6724506	No
Error	2097	108540.6608756	51.7599718			
Total	2104	109316.8762629				

Step 5 : Residual Analysis & Assumption's checks

Choose

Statistics → Analysis of (Co)Variance → ANCOVA

and choose

Residual Analysis & Assumption's Checks.

You get the following window:

Residual Analysis & Assumption's Checks

? X

Normality of Residuals

Statistical Tests

- ☐ Shapiro-Wilk Test
- ☐ Anderson-Darling Test
- ☐ Kolmogorov-Smirnov Test

Plots

- ☐ Normal Q-Q Plots
- ☐ Histogram of Residuals
- ☐ Density Plot of Residuals

Homogeneity of Variance (Homoscedasticity)

Statistical Tests

- ☐ Levene's Test
- ☐ Brown-Forsythe Test
- ☐ Bartlett's Test (for normal data)

Plots

- ☐ Residuals vs Fitted Values
- ☐ Residuals vs Dependent Variable
- ☐ Boxplots by Group
- ☐ Violin Plot by Group

Residual Diagnostics

Plots

- ☐ Residuals vs Observation Order

Model Adequacy

Plots

- ☐ Residuals vs Fitted values
- ☐ Residuals vs Each Factors
- ☐ Residuals vs Covariates

then choose the tests you want. For example, choose the following:

Residual Analysis & Assumption's Checks

?

X

Normality of Residuals

Statistical Tests

- ☒ Shapiro-Wilk Test
- ☒ Anderson-Darling Test
- ☒ Kolmogorov-Smirnov Test

Plots

- ☒ Normal Q-Q Plots
- ☒ Histogram of Residuals
- ☒ Density Plot of Residuals

Homogeneity of Variance
(Homoscedasticity)

Statistical Tests

- ☒ Levene's Test
- ☒ Brown-Forsythe Test
- ☒ Bartlett's Test (for normal data)

Plots

- ☒ Residuals vs Fitted Values
- ☒ Residuals vs Dependent Variable
- ☒ Boxplots by Group
- ☒ Violin Plot by Group

Residual Diagnostics

Plots

- ☒ Residuals vs Observation Order

Model Adequacy

Plots

- ☒ Residuals vs Fitted values
- ☒ Residuals vs Each Factors
- ☒ Residuals vs Covariates

Save

Cancel

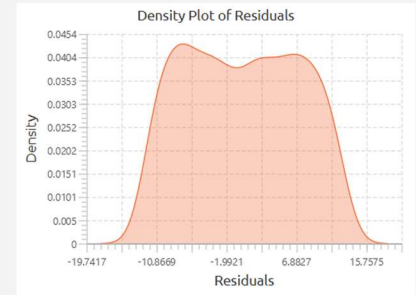
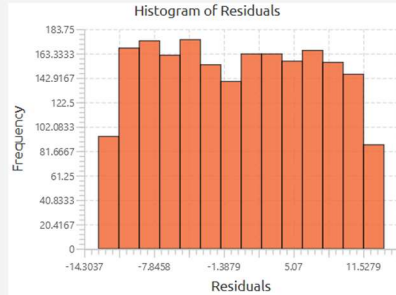
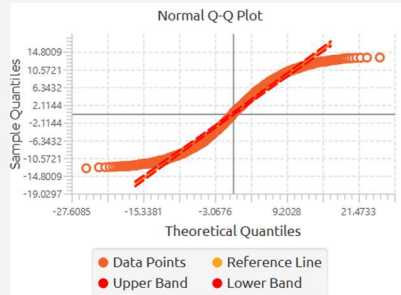
After that, click execute and you get the results (some of them below):

ANCOVA

Normality of Residuals

Statistical Tests

Normality Test	Statistic	Prob	P-value	Decision
Shapiro Wilk	0.957	1.000	0.000	Reject Null Hypothesis
Anderson Darling	23.840	1.000	0.000	Reject Null Hypothesis
Kolmogorov Smirnov	0.065	1.000	0.000	Reject Null Hypothesis



Homogeneity of Variance

Statistical Test (Univariate)

Test	F-statistic	df1	df2	P-value	Decision
Levene	0.198	3	2101	0.898	Fail to Reject Null Hypothesis
Brown - Forsythe	0.227	3	2101	0.878	Fail to Reject Null Hypothesis
Bartlett	0.245	3		0.970	Fail to Reject Null Hypothesis

